Defending Machine-Learning Intrusion Detectors Against Data Poisoning Attacks

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Abstract
Intrusion detectors can train themselves to ignore malicious traffic if their labels are corrupted. Label contamination is a major security threat for Internet of Things (IoT) network classification systems. In this research, we examine the influence of label-flipping poisoning attacks on the performance of Random Forest and Logistic Regression classifiers based on CICIoT2023. The following experiments will compare random label flipping of attack samples with targeted poisoning of DDoS attack samples at 5%, 10% and 20% of attack data volume ratios. A two-stage defence against label flips uses k-nearest-neighbour label consistency checking and class-based statistics-driven outlier elimination, which does not require clean training data to work. Evaluation will take into account performance under clean, poisoned and sanitized training data with the same test data for all three conditions. Detection metrics to be compared include accuracy, precision, recall, F1-score and false positive rate, where attack-category recall can reveal the problem that accuracy fails to detect. Poisoned sample detection and removal of correctly labelled data will be used to measure the defence performance. Experiments will be carried out on five different random seeds and also with equal numbers of flipped labels for comparison.

I. Introduction

A. Background: ML-Based NIDS

The increasing use of Internet of Things (IoT) devices has created a greater need to monitor connected environments for malicious activity. IoT cameras, sensors and other smart devices generate network traffic that can be analysed to identify potential attacks. Machine learning based network intrusion detection systems (NIDS) use labelled traffic samples to learn patterns that distinguish benign activity from attacks. CICIoT2023 provides a benchmark dataset for studying these systems, containing traffic collected from 105 IoT devices and covering 33 attack types [1].

The effectiveness of supervised detection depends partly on the quality of the training labels. Label-flipping data poisoning attacks exploit this dependence by corrupting labels without changing the traffic features [2]. The erroneous label may cause the model to learn the wrong label for the targeted sample [2]. As a result, similar attacks may become more difficult to detect. This project examines the effects of label corruption on IoT intrusion detection and explores whether data sanitisation can identify suspicious training samples and reduce their impact before model training.

B. Problem Statement

The ability of a network intrusion detection system based on machine learning to distinguish between normal and attack traffic depends *partly* on its labelled training data. In case the labelling is poisoned, the system might learn erroneous relationships and thus decrease its capacity to detect attacks. Research in literature reveals that label-flipping attacks can lower the performance of deep learning-based network intrusion detection systems [3]. Further research needs to be conducted to see how such attacks influence the models and dataset used in this research.

The aim of this research is to assess the influence of random and targeted label-flipping attacks on Random Forest and Logistic Regression models trained on CICIoT2023.

Moreover, this research aims to explore if checking the consistency of the neighbours' labels and outliers can help detect the abnormal training samples.

C. Motivation

As of the end of 2024 the number of connected IoT devices stood at 18.5 billion, and is projected to reach 39 billion by 2030 [4]. The majority of these devices, including cameras, sensors, and smart-home hubs, have limited processing power and cannot run traditional host-based security software. Therefore, network-level intrusion detection has been the main line of defense, and machine learning (ML) models trained on a set of data like CICIoT2023 have been reported to reach high accuracy in detecting intrusions [1].

Such models, however, are as accurate as the labels they learn from. Labels for training are frequently generated by automated tools, third-party data and feeds, or shared threat data, and all of these can be modified. NIST states that data poisoning is one of the important classes of attacks against ML systems [2]. One of the most basic and common methods of poisoning is to flip the label. No part of the network traffic is modified by the attacker, just its labels. The damage is also stealthy: the model appears to train normally and can keep high overall accuracy while learning to ignore certain attacks. A targeted attacker can focus this effect on a particular attack type (such as DDoS) and leave a hole in the detector. This gives rise to a real-world need to quantify the impact of label flipping and test whether lightweight data-sanitisation checks can identify and remove poisoned labels before training. This project addresses both questions.

D. Scope and Objectives

This project is mainly related to the label flipping poisoning attacks in ML based network intrusion detection for IoT. The experiments are done with the CICIoT2023 dataset [1] that has been limited to a binary classification problem (Benign vs. Attack). To ensure manageable training times, a stratified sample of the data set is used.

Two attack strategies are tested. In random flipping, samples are flipped from any attack class. In targeted flipping, only DDoS samples are relabelled as Benign. All strategies are tested with 5%, 10% and 20% poisoning, and each run starts from clean data. The performance of two classifiers is compared, namely Random Forest and Logistic Regression. There are two pre-training defences that are applied and tested: one is a k-nearest-neighbour label consistency check, and the other is statistical outlier filtering. The three conditions were tested to compare each configuration: clean baseline, poisoned without defence, and poisoned with defence. The accuracy,precision, recall, F1-score, false positive rate and attack-category recall are used to measure results.

This project does not include poisoning attacks affecting traffic characteristics, deep learning models, multi-class detection, federated learning settings and live network deployment. All experiments are performed in an offline manner, using a publicly available dataset and no real networks or devices are attacked.

E. Relevance to Real-World Cybersecurity Practice

Machine learning is being increasingly studied for intrusion detection in IoT networks, where a massive amount and variety of network traffic make automated intrusion detection useful. Reliability of AI-powered intrusion detection systems (IDSs) is thus a crucial cybersecurity issue. These systems increase security threats as well, since they are dependent on the quality and integrity of the training data [5].

In the National Institute of Standards and Technology (NIST) list of adversarial machine learning threats, data poisoning is the situation where the model's behaviour and reliability can be

impacted by manipulated training data. These attacks can happen without being noticed until the model fails to make the correct predictions in use [2].

This is especially applicable to ML-driven Network Intrusion Detection Systems (NIDS).In a label-flipping attack, the attacker flips the training labels (e.g., a malicious network traffic is given a benign label). The model can then learn false relationships between the flow of data on the network and its security level, making it less capable of detecting future attacks when deployed. Thus, this work focuses on the effect of data poisoning due to label flipping on IoT NIDS and examines if it is possible to identify and decrease the impact of the corrupted training labels by implementing a mitigation approach.

I. Literature Review

A. Overview of ML-Based Intrusion Detection

Intrusion detection systems (IDS) are designed to detect intrusion and are typically divided into two categories: signature-based and anomaly-based [6]. A signature-based system matches traffic to known attack signatures. They work for known threats but not for new and zero day attacks. Anomaly-based systems learn from the normal behaviour and identify any deviations. This enables them to see attacks that they have not encountered before but results in a higher number of false positives [6].

Machine learning (ML) is currently popular in constructing network intrusion detection systems (NIDS). Using features extracted from network flows, rather than hand-written rules, ML models learn to distinguish between benign and malicious traffic [7]. For the IoT applications, both classical ML models and deep learning models are well-studied, and recent studies reviewed also address the training and deployment of these models [5].

An ML-based NIDS is dependent on the quality of its training data in large part. Older benchmark datasets like NSL-KDD have been found to be inadequate to represent the real-world IoT network traffic [6]. This is addressed by newer datasets like CICIoT2023 which registers the traffic of 105 real IoT devices for 33 attack types and models trained on CICIoT2023 attain high detection performance [1]. Most of these models, however, are supervised, meaning that each training example must have a correct label. The cost of producing these labels is high, and manual verification is necessary in most cases [7]. Because the model fully trusts these labels, the integrity of the training data becomes a security concern. If the attacker can alter the labels, then the detector can learn the wrong things, resulting in so-called data poisoning attacks, which are discussed next.

B. Data Poisoning Attacks

Data poisoning is an attack conducted during training, where the means of attack involve poisoning the training data to alter the behaviour of a machine learning model[2]. NIST describes poisoning attacks based on the attacker's goal: availability, targeted, backdoor and model poisoning. Work on poisoning has been explored in the cyber security field, including in the generation of worm signatures and spam filtering systems, where it is shown that systems that learn from newly arrived data can be impacted by data poisoning[2].

A form of poisoning that is relevant for this project is called label flipping and occurs when the correct label assigned to a training example is changed by an attacker while not necessarily changing the feature values of that example. NIST considers label manipulation a possible poisoning attack, while Chang et al.[8] show that the effectiveness of label-flipping attacks is dependent on the choice of poisoned samples. They find that, for 10% poisoning rate, the accuracy of the neural network on the test set was degraded by around **1%** in the case of random label flipping, and by up to **11.6%** on individual datasets when the optimised

attack was used. This implies that the effectiveness of the poisoning will depend on the percentage of the contaminated samples, as well as the strategy and knowledge of the attacker. So, this project will focus on the evaluation of the effect of the different selection strategies (random and targeted) on an IoT NIDS with the investigation of label flipping with different poisoning rates.

C. Poisoning in Security and NIDS Contexts

Label poisoning poses a significant threat to security systems because a compromised detector could continue to operate without any visible sign of failure and misclassify dangerous behavior. Wulnye et al. [9] used three IoT datasets, one of which is CICIoT2023, tested against four classifiers and four poisoning methods, one of which is label flipping. What they found was that the accuracy of their results decreased as much as **40%** in worst case scenarios due to the negative impact of label flipping in all datasets. Random Forest and Gradient Boosting maintained their performance better than Logistic Regression and deep neural networks. However, the study only focused on the impact of poisons on models. It did not try any defence against it.

Other work has investigated methods of defending against these attacks in security related classification problems. Nowroozi et al.[10] attacked a malicious-URL classifier with poising rate of 2%-5%.The attack succeeded in 50-65% of attempts,yet accuracy dropped by no more than five percentage points, This implies that general performance metrics can allow such attacks to go unnoticed. The authors also used a k-nearest-neighbour method to fix labels but that method needed access to the original trusted dataset, which is not always possible.

These two studies demonstrate that label flipping can have a significant impact on security machine learning systems and that it can be challenging to detect label flipping by observing the accuracy of a machine learning model. The studies reviewed have not collectively investigated random and targeted label flipping at different poisoning rates in an IoT Network Intrusion Detection System (NIDS) while evaluating methods to reduce the damage. This is the missing piece that helps make the design of this project move forward.

D. Existing Defenses

Methods for preventing data poisoning include the use of data sanitization techniques prior to training and making the training process more robust to poisoned data points [2]. One example of a data sanitization technique involves partitioning of new data into smaller sets, inspecting them using a clean deep learning model, and filtering out any poisoned sets detected [11]. In this way, new data can be inspected before retraining takes place. However, it needs a clean inspection model, thus limiting its practicality based on the presence of a clean model.

NIST considers label-cleaning techniques used to prevent label flipping attacks, outlier detection and clustering algorithms for poisoned data identification [2].

Defense mechanisms can be applied to collaborative training as well. One mechanism analyzes gradients of the output layer and applies clustering to filter updates by malicious participants [12]. The evaluations of that mechanism involve intrusion detection dataset such as NSL-KDD and CICIDS-2017. As opposed to data sanitisation, which removes or corrects suspicious training samples, this defense mechanism filters model updates. Yet, it uses a federated learning framework as opposed to the central training framework of this project, and therefore its results cannot be taken as evidence of effectiveness for the chosen models.

The above-stated defenses show how these techniques were used on training samples or model updates. This project will investigate outlier filtering and neighbor label consistency checks before training of Random Forest and Logistic Regression on CICIoT2023 dataset.

Comparison of Existing Work

TABLE I
COMPARISON OF EXISTING WORK

Ref.	Dataset	Attack / Defense	Key Finding
[3]	CIC-IDS 2017, CSE-CIC -IDS2018	Label flipping vs. deep-learnin g NIDS; no defence	Accuracy fell from ~97% to 29% as flipped labels increased
[8]	General tabular datasets	Fast optimised label-flippin g attack; no defence	At 10% poisoning, random flips cost ~1% accuracy; optimised attack up to 11.6%.
[9]	Four IoT datasets incl. CICIoT2 023	Four poisoning methods incl. label flipping; impact only	Accuracy dropped up to 40%; RF and GB more robust than LR and DNN.
[10]	Malicious URL data	Label flipping (2-5%); ensemble trees with k-NN label fixing	Attack succeeded in 50-65% of cases with ≤5 percentage-po int drop in training accuracy.; k-NN fix needed trusted data.
[11]	Deep-lear ning (transfer-l earning) data	Poisoning; dataset partition and inspection defence	Removes poisoned sub-datasets, but requires a clean inspection model.
[12]	NSL-KD D, CICIDS-2017	Multi-label poisoning; gradient clustering in federated learning	Filters malicious updates; federated setting, not central training.

Research Gap

The studies reviewed reveal the risks of label flipping in ML security, but several questions remain unanswered.

First, the damage to be caused by the flipping of labels will depend on which samples are flipped. Chang et al. [8] discovered that random flipping caused little loss, but a carefully chosen attack caused significant losses. Their work, however, was based on general tabular data, rather than on network traffic. Second, intrusion detection studies have been focused on measuring the damages but have not tested the defence. Mohammadian et al. [3] found that the accuracy of a deep learning NIDS decreased from nearly 97% to 29% as the number of flipped labels increased. Their study used non-IoT datasets, i.e. CIC-IDS2017 and CSE-CIC-IDS2018, and they did not consider any defence. Wulnye et al. [9] reported the same damage for the CICIoT2023 and other IoT datasets, and observed that Random Forest was more resilient than Logistic Regression; however, they did not go beyond measuring impact. Third, much of the defence research focussed on defence of image data [13] and some sanitisation methods need a clean inspection model [11]. The malicious-URL study [10] imposed a similar requirement for trusted data, which is typically not possible in practice.

Based on the studies reviewed, limited research has investigated all these aspects together using IoT network traffic. In particular, none of the reviewed studies has compared random label flipping with targeted label flipping at various poisoning rates on CICIoT2023 and then investigated if any lightweight sanitisation defenses, that do not require trusted data, can detect and repair the damage. This project aims to rectify that by quantifying the effects of each

attack strategy and recovery provided by K-NN label checking and outlier filtering over both Random Forest and Logistic Regression models.

II. Methodology

A. Dataset and Preprocessing

This study uses the CICIoT2023 dataset [1], a labelled IoT network traffic dataset that was built from **105** IoT devices, **33** attack types and benign traffic. The attacks are organised into seven categories: DDoS, DoS, Reconnaissance, Web-based, Brute Force, Spoofing and Mirai. The dataset offers **46** extracted network traffic features, calculated as averages over fixed windows of **10** or **100** packets depending on the attack type.

The original dataset is very large with about **46.7 million** rows and is highly imbalanced.**DDoS** accounts for about **73%** of all records , Benign traffic for about **2.4%.** Brute Force is the smallest attack category, with only 13,064 records. To keep experiments manageable, a balanced subset was drawn from the original files by random sampling(seed 42) [14]. With each of the seven Attack categories contributing about 13,000 records.There are no duplicated or synthetic rows in the data. This gives a total of **91,000 Attack records.** The same number of Benign records **91,000** is sampled. Therefore Benign and Attack are balanced in the binary task. The dataset contains **182,000 records** in total. No duplicate feature vectors were found in the subset. Because the subset is balanced by design its class proportions do not reflect traffic so the results show the relative effect of poisoning rather than absolute deployment performance. This makes repeated experiments practical on computing resources while keeping all seven Attack categories.

The task is formulated as binary classification: Benign (label 0) and Attack (label 1 which combines all **33** attack types). The attack-category column is used to derive the label and to select samples for targeted poisoning (Section III-B). It is removed, together with the original fine-grained label, before model training so it is never used as a feature.

The preprocessing steps in order are:

- Remove rows with invalid values and any duplicate rows. This is done before splitting so that identical rows cannot appear in both the training and test sets.
- Split the data into training and test sets using a stratified **80/20** split by category with a fixed random seed giving **145,600 training** and **36,400 test records.**
- Standardise the features with StandardScaler fitted on the training data then apply it to the test data.

This prevents information from the test set from influencing preprocessing. Label poisoning is applied to the training set only. The test set always keeps its clean labels.

B. Threat Model and Attack Design

The threat model assumes an attacker who compromises the training data of a NIDS by changing labels while leaving the network traffic features untouched. In practice labels may come from analyst annotations, automated labelling tools, or Shared threat-intelligence feeds, all of which can be corrupted (Section I-C). The attacker's goal is to have traffic learned as benign. The attacker can alter the labels of a fraction of the training samples, but cannot modify feature values the test data or the training algorithm. The attacker knows the task and the data structure but has as no access to model parameters, which makes this a black-box poisoning scenario [2].

Two Attack→Benign label-flipping strategies are used at poisoning rates of 5 %, 10 % and 20 %:

Random flipping: attack samples are chosen at random from all 72,800 training attack samples and relabelled as benign. The rate is the proportion of training attack samples that are flipped.

Targeted flipping: samples from one attack category for example DDoS are relabelled as benign. Each category has 10,400 training samples. The rate is the proportion of that category's samples that are flipped.

Because the two strategies draw from pools of different sizes, the same rate flips a very different number of labels. At 10 % for example random flipping changes 7,280 labels while targeted flipping changes 1,040. We therefore report the number of flipped labels for every run. We also run random flipping with the same number of labels as targeted flipping (520, 1,040 and 2,080) so the two strategies are compared on an equal footing. Each run begins from a copy of the clean training set so poisoning never builds up. Sample selection uses fixed seeds. Every experiment is repeated over five seeds (42–46) to give the mean and variability.

C. Detection Models

The classifiers trained are Random Forest, an ensemble that is non-linear, and Logistic Regression, the linear base model. Random Forest is selected because the ensembles of trees are shown to be performing well on tabular network-flow features and relative performance of ensembles of trees is relatively stable in the presence of data poisoning [9]. Logistic Regression is considered as a simple and interpretable baseline model which has been shown to be more sensitive to poisoning [9]. The results indicate if there is a difference in the amount of damage from flipping labels between the two models.

Random Forest uses 100 trees, no maximum depth and the Gini criterion. Logistic Regression is applied to the standardised features with L2 regularisation (C = 1.0), the lbfgs solver and a max of 1,000 iterations. Both models are set to a fixed random seed. Since the same hyperparameters are used in all of the conditions (clean, poisoned and sanitised), all differences in performance are from the training data only.

D. Proposed Defense

The defence sanitises the training set before the models are trained, using only the possibly poisoned data. Unlike the k-NN defence in [10], it does not require a trusted clean dataset. It has two stages.

1) k-NN label check: For each training sample, its k = 10 nearest neighbours are found in the standardised feature space using Euclidean distance. If 7 or more neighbours have a different label, the sample is marked as a possible flip and is excluded. The idea is that a DDoS sample relabelled as Benign still looks like DDoS, so its neighbours will mostly have the Attack label.

2) Outlier filtering: Outlier detection is a typical data-sanitisation defence against poisoning [13]. Once the k-NN check is done, the centroid of each class is calculated and the distance of each sample to the centroid of the class it claims to belong to. Any samples that fall more than three standard deviations from the mean for the class are considered statistical outliers and excluded from the data set.

Flagged samples are not relabelled, but removed. Removal is the more conservative choice, because a wrong correction would itself add label noise.

The direct assessment of the defence is also done because the experiments will show which ones were flipped or not flipped. Precision is the share of removed samples that were actually poisoned, and recall is the share of poisoned samples that were caught. All thresholds are pre-set and never tuned on the test set.

E. Evaluation Plan

Random Forest and Logistic Regression will be tested under three scenarios: training using clean data, training using poisoned data without defence, and training using sanitized poisoned data. In all cases, the same train/test split and unmodified test data will be used. The hyperparameters of models will be kept constant for all experiments, while each poisoning scenario will begin using a fresh copy of the clean training data.

The poisoning rate of 5%, 10%, and 20% will be used for the samples that could be flipped. Random flipping will pick up the samples from the whole population of attack samples in the training set, whereas targeted flipping will pick up samples from a selected attack class, for example, DDoS. The Attack label will be changed to Benign, and the Benign samples will not be changed. Because the size of the eligible population varies, the number of flipped samples and the fraction of the total training set will be reported together with each poisoning rate.

Each experiment will be repeated five times using random seeds ranging from 42 to 46. The seed for splitting the dataset into training and test sets will be kept constant, but seeds used for the selection of poisoned points and the randomness of models when required will be changed between runs. Conditions will be tested with the same seed and poisoning selection in order to facilitate comparisons. Results will be reported as mean ± standard deviation across the five runs.

Attack detection will be evaluated by means of accuracy, precision, recall, F1-score, and false positive rate. In particular, it is important to pay attention to the ability of detection to identify attacks and to the performance of attack recall within specific attack categories. Comparative analysis will evaluate how much poisonings decrease attack detection performance and how much sanitization improves it. Comparison between attack strategies will take into account not only equal rates but also the number of flips.

Defense performance will additionally be assessed through the proportion of poisoned samples detected, the proportion of removed samples that were genuinely poisoned, and the proportion of correctly labelled samples removed. Original labels and poisoning records will be used only for subsequent evaluation and will not be available to the defense during filtering.

F. Tools and technologies

Experiments will be implemented in Python. Pandas and NumPy will be used for data handling, scikit-learn for preprocessing, the random forest and logistic regression classifiers and k-nearest-neighbour search, and Matplotlib for visualising results.

G. Ethical and Legal Considerations

Offline experiments will be run using the CICIoT2023 data set. Attacks based on label flipping will be carried out on local copies of the training data and no such attacks will be done on any live networks, IoT devices or third party systems. Original data set will be saved and experiment details will be noted down for future reference purposes.

Before using or redistributing the data set, its terms of access and usage will be examined. All files within the data set will be checked for any personal information and no such data will be included in the repository. All examples will make use of either synthetic or anonymised data. The repository will include the code, configurations, documentation and experimental results.

The project will be carried out adhering to all research and academic integrity policies of the university. Before doing any data collection, distribution or working with an external system, all permissions and legal requirements with regard to the data set will be looked into. AI will be used as per the policy of the module and all AI-assisted content will be reviewed and verified by the team. Results will be reported honestly, including unsuccessful defenses and limitations.

H. conclusion

In this report, the plan for a study of label-flipping data poisoning attacks on ML-based intrusion detection in IoT networks has been presented. The literature review revealed that “label flipping” can seriously weaken an intrusion detection system while its overall accuracy still looks normal.It additionally found that current research has primarily been conducted on measuring damage, but not on testing practical defenses. Proposed defences are frequently based on trusted clean data, something that is seldom available in real deployments.

The project will tackle this issue by flipping Attack labels to Benign at random and targeted attacks at 5%, 10% and 20% poisoning rates in CICIoT2023. It will measure the effect on Random Forest and Logistic Regression classifiers. It will then evaluate a two-stage sanitisation defence, which includes a k-nearest-neighbour label check and statistical outlier filtering, that assumes no trusted data. The performance of each configuration will be measured with five random seeds using clean, poisoned and sanitised conditions with accuracy, precision, recall, F1-score and false positive rate.

The desired result is a clear understanding of the damage label flipping causes to IoT intrusion detection and whether lightweight data sanitisation can reverse it. This design will be implemented and evaluated in the next phase.

III. Team Plan

Table II lists the planned tasks and owners.

TABLE II
Team Plan

Task	**Owner**	**Timeline**
Introduction: Background (ML-Based NIDS) ,Abstract	Neam	Done,week 1
Introduction: Problem Statement	Neam	Done,week 1
Introduction: Motivation	Haneen]	week 1
Introduction: Scope and Objectives	Haneen	Done,week 1
Introduction: Relevance to Real-World Practice	Mariam M	Done,week 1
Literature Review: Overview of ML-Based IDS	Haneen	Done,week 3
Literature Review: Data Poisoning Attacks	Mariam M	Done,week 3
Literature Review: Poisoning in Security and NIDS	Mariam M	Done ,week 3
Literature Review: Existing Defenses	Neam	Done ,week 3
Literature Review: Comparison of Existing Work	All	Done week 4
Literature Review: Research Gap	Haneen	Done,week 3
Methodology: Dataset and Preprocessing	Mariam M	Done,week 4
Methodology: Threat Model and Attack Design	Mariam M	Done,week 4
Methodology: Detection Models	Haneen	Done,week 4
Methodology: Proposed Defense	Haneen	Done,week 4
Methodology: Evaluation Plan	Neam	Done,week 4
Methodology: Ethical and Legal Considerations	Neam	Done,week 4

Table III
Plan For Weeks 4-8

Task	Owner	Week
Repository setup (README, project structure, issue board), preprocessing pipeline and stratified sampling	Mariam M	4-5
Clean baselines (Random Forest, Logistic Regression)	Haneen	5
Random and targeted label-flipping implementation	Mariam M	5-6
k-NN consistency check and outlier	Haneen	6

filtering		
Evaluation metrics and defence-assessment code	Neam	6
Full experiments (5 seeds, 3 poisoning rates, equal-flip comparison)	All	6-7
Results analysis, tables and figures	Neam	7
Deliverable 2 writing, team review and submission	All	7-8

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